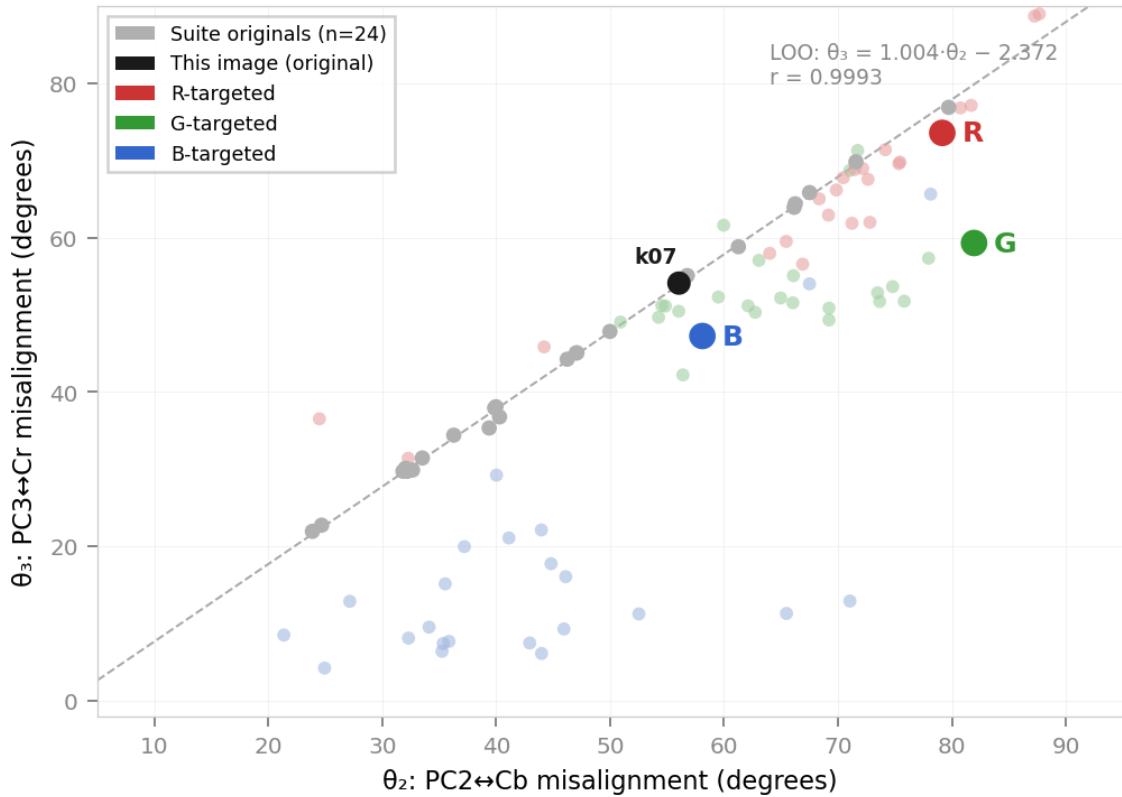


Compression Profile: KODIM07 768x512 | 24-bit RGB | PNG (lossless)

shuttered windows | Weakly One-Dimensional (PC1 = 88.77%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8330	0.7634	0.9084	0.835	56.04°	54.13°	+0.24°
R-targeted	-0.2905	-0.2359	0.9437	0.490	79.16°	73.61°	-3.49°
G-targeted	-0.1207	0.9897	-0.1033	0.405	81.95°	59.33°	-20.58°
B-targeted	0.9635	-0.0412	0.1070	0.371	58.1°	47.27°	-8.69°

R-targeted isolates red (R-G, R-B < 0.29) while preserving G-B coupling at 0.944.
G-targeted isolates green (R-G, G-B < 0.12) while preserving R-B at 0.990.
B-targeted preserves R-G at 0.964 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM07768x512 | 24-bit RGB | PNG (lossless)

shuttered windows | Weakly One-Dimensional (PC1 = 88.77%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

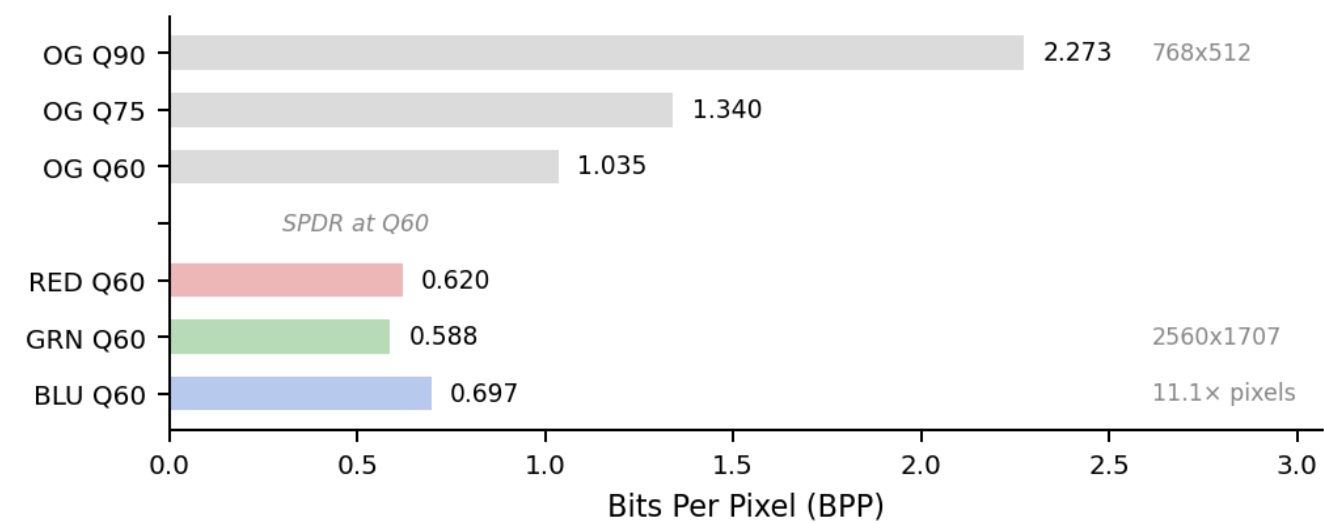
ORIGINAL (kodim07)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	553.0 KB	11.522	393,216
JPEG Q90	JPEG	768x512	109.1 KB	2.273	393,216
JPEG Q75	JPEG	768x512	64.3 KB	1.340	393,216
JPEG Q60	JPEG	768x512	49.7 KB	1.035	393,216

REDISTRIBUTED (kodim07 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	330.6 KB	0.620	4,369,920
GRN Q60	JPEG	2560x1707	313.9 KB	0.588	4,369,920
BLU Q60	JPEG	2560x1707	371.9 KB	0.697	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM07768x512 | 24-bit
RGB | PNG (lossless)

shuttered windows | Weakly One-Dimensional (PC1 = 88.77%)

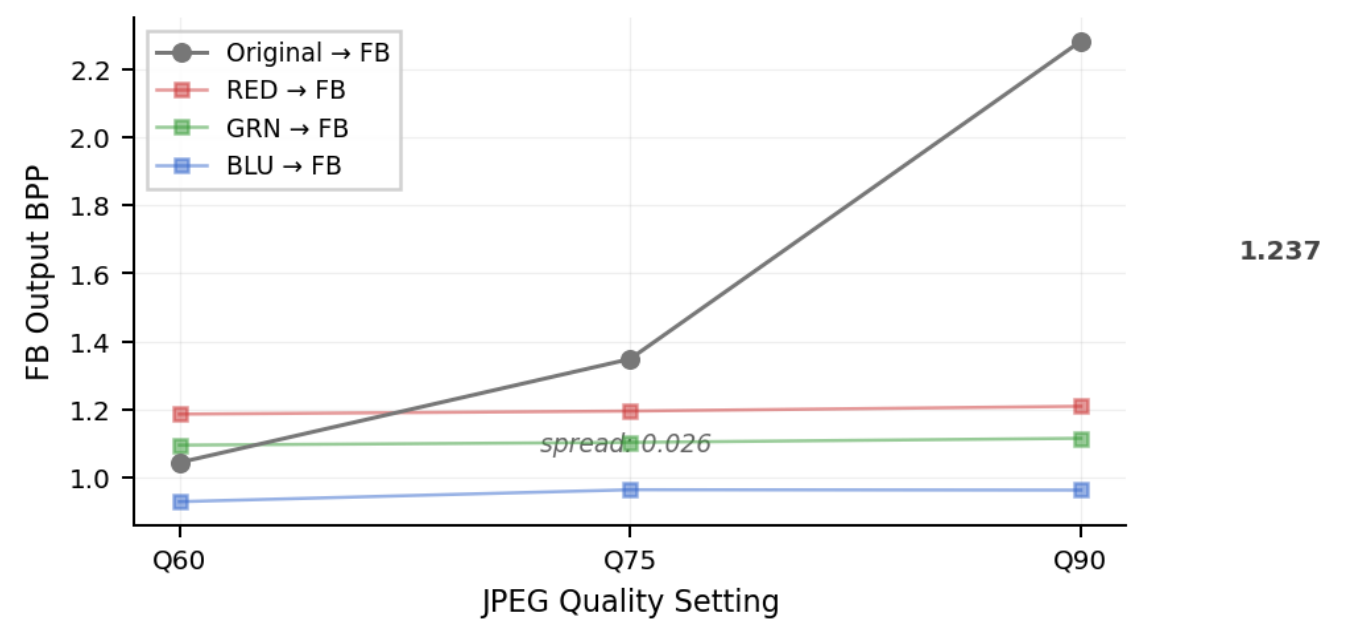
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.067	–
R-targeted → FB	2560x1707	2048x1366	1.210	41.5%
G-targeted → FB	2560x1707	2048x1366	1.116	46.0%
B-targeted → FB	2560x1707	2048x1366	0.964	53.4%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.127	45.5%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8340 vs 0.835 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.237 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.026 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim07

Best BPP (Q60, direct)	0.588	2560x1707, 4,369,920 px
Best BPP through FB	0.964	2048x1366 output
vs OG Q90 → FB	57.8%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216